

WHEN AI GETS IT WRONG: CONSCIOUS COMPETENCE DEVELOPMENT THROUGH STUDENT DESIGN REFLECTIONS

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Abstract – *Reflections in engineering are a fruitful tool for encouraging lifelong learning and provide instructors with insight into how to adapt the learning environment to meet students' needs. However, reflections are impractical for large classes. This research explores the use of a latent Dirichlet allocation model to automatically classify students' design reflections, thereby improving the efficacy of their reviews.*

The model showed promising quantitative performance, but from a human reviewer's perspective, it lacked interpretability. The contradicting results encouraged a reflective discussion of the potential misuses and risks of artificial intelligence (AI), both within and outside the classroom, without a qualitative review. Based on the authors' experiences and lessons from this study, this paper advocates for professional engineering licensure bodies to modernize policies to encourage thoughtful, rigorous evaluation of AI models before deployment. This article was written to encourage engineering educators to utilize AI critically and responsibly.

Paper Type: Research

Keywords: Critical reflection, professional practice, design projects, instructional change, pedagogical content knowledge

1. INTRODUCTION

In engineering design courses, reflections help students solidify their learning and strengthen their strategic thinking, while also serving as valuable teaching aids that increase student engagement and encourage lifelong learning [1,2,3]. As class sizes increase, it becomes cumbersome to manage, interpret, and provide timely feedback to students' reflections while minimizing bias in their interpretations. As a result, the pedagogical value of reflections is often not fully realized [4]. One potential solution to this problem is to utilize artificial intelligence

(AI) to enhance the efficiency of reviewing student design reflections.

The adoption of AI has accelerated across numerous domains, promoting widespread integration into engineering pedagogical practices [4]. Educators have employed AI for purposes such as providing automated feedback, predicting student performance, and assignment curation. Large language models (LLMs), such as ChatGPT and Gemini, are frequently chosen to support instructional activities due to their versatility and reasoning capabilities [5]. Alongside the widespread uptake of AI, there has been a proliferation of resources aimed at teaching others how to train and deploy AI models themselves [6]. Naturally, many educators and professionals have begun to experiment with training their own models for their own distinctive purposes.

Likewise, the initial aim of this research was to train an AI model to automatically extract recurrent and key themes from engineering student design reflections. In large classes, such a model could have broad implications, providing insight into team dynamics, project progress, and concepts that require further reinforcement, thereby enabling timely and targeted instructional interventions. This article incorporates both the original research objective and a critique on the use of AI in educational settings in response to the research findings. The research objectives are:

(1) Train an AI model to analyze the rich content from student design reflections to characterize how key themes progress throughout design courses.

(2) Identify a framework that promotes effective and ruminative use of AI in educational settings through the lessons learned from the implementation and evaluation of the AI model.

In the first objective, an AI model is applied to extract recurrent themes from engineering student reflections. The model's result demonstrated how even with reasonable and careful rigour in the selection of technologies, it is easy to have application of an AI system go awry and produce incorrect or invalid results. In particular, the results from

the first objective demonstrate how analyses of the reflections from an AI model that look entirely reasonable on first encounter may not be even remotely similar to how an instructor may interpret and mark such material. The process and results of the model inspired discussion, leading to the conceptualization of the second objective and the identification of a methodological framework for the use of AI in engineering education. In the discussion, a list of considerations is presented that engineering educators should reflect upon before integrating AI into their pedagogical workflows. The considerations are framed around the concept of the Hierarchy of Competence, and opinions are provided on how it should be adapted in AI-enabled engineering pedagogy.

2. BACKGROUND

Self-reflection has many proven benefits across an array of domains within engineering education, most notably to promote life-long learning. Students have reported that frequent reflection in engineering design settings helps them stay on track with deliverables, plan ahead, and identify strengths and knowledge gaps [7]. Furthermore, research has shown that reflections help students find and understand their role within teams, allowing them to contribute more effectively [8]. Another study has shown that reflections can improve students' self-confidence in course content and encourage them to ask more pertinent questions to challenge their understanding [9].

Recurring themes that emerge from student reflections can provide insights for instructors into common areas of misunderstanding and class learning outcomes. In engineering design settings, reflections can reveal important characteristics of team dynamics and can even prompt targeted intervention from the instructor if necessary [10]. Reflection themes can also elicit tailored lesson plans to support the specific needs of each students' project. These benefits are often not realized in large classrooms due to the monotonous and time-consuming nature of reviewing corpora.

Several studies have outlined (and promoted) use cases of AI and LLMs for student reflection review in large classrooms [11, 12, 13]. Perhaps the most common application is the use of LLMs to generate supporting text describing the content of student reflections to provide the grader with interpretations of the students' work to improve review efficiency [14, 15]. Other studies have explored summarizing large documents and assigning specificity scores, where reflections containing more detailed and specific content are considered to be of higher quality [16]. One article even suggests that AI models, including vision systems, could be used to automate administrative tasks, such as attendance, to leave the educators with additional time to focus on delivering course content and providing

feedback [17]. One paradigm that emerged during the literature review is that most frameworks focus on assigning predefined scores or summarizing bodies of text, rather than processes designed to identify classroom wide patterns and themes within student reflections.

The aim of the first research objective was to design an AI powered process to extract recurring themes from student reflections. Although LLMs are powerful, there exist several key limitations that render them less attractive for this purpose and an alternative was pursued. For example, many LLMs lack contextual understanding and cannot easily cross-reference multiple text documents at once [18]. In design environments where students are arranged into teams, these limitations hinder the use of LLMs to extract valuable information from reflections across teams and individuals. Furthermore, LLMs may struggle to infer latent thematic elements within a design team, such as underlying team dynamics, due to their limited cross-referencing capabilities [19].

As an alternative to LLMs, latent Dirichlet Allocation (LDA) topic modelling was applied as a prospective method to study student reflections due to its interpretability and simplicity, which enable educators to train their own models that capture contextual understanding within design projects [18]. Other methods to model student design reflections include, but are not limited to, topic modelling using bidirectional encoder representations from transformers (BERT), applying data preprocessing before reasoning with an LLM, and conventional techniques that involve manually encoding and classifying text based on similarities [20].

An LDA model is an unsupervised probabilistic model technique used to discover hidden thematic elements in a corpus of text [21]. Topics are represented as collections of words that are likely to be related; a human can then assign encompassing labels to each of these groups. For instance, in the excerpt below, the underlying topics may be "teamwork", "challenges," or "communication."

"Our team communicated well, but it was hard to collaborate on the final project."

In more complex sentence structures, themes may be more difficult to infer for humans. It is also important to note that a key limitation of LDA models is that they are unable to extract words that were not present in the training data (without pretraining) [22]. During training, the developer specifies the number of topics for the model to search for in the documents.

If the model were to be successful, an educator could upload reflections from a large class (>75 students), and the model could identify emergent themes for the group while accounting for how themes developed from previous weeks. Although, at the time this paper was written, limited cross-referencing capabilities made LLMs less practical than alternatives, the field of AI is rapidly advancing, and

it could soon be possible for data from previous weeks to influence themes.

3. FIRST RESEARCH OBJECTIVE: METHODOLOGY

3.1 Study Design

Towards the first research objective, an LDA model was trained on weekly reflections from 83 second-year design students [20]. Each participant completed between 14 and 19 reflections, totalling 1372 reflections available for model training. Data was collected over the course of two second-year design cohorts offered in 2021 and 2022. Reflections were submitted weekly and spanned two team-based design projects. The first project task was common for all teams to solve a problem in a local community and was five weeks long. New teams were formed for the second project. Unique topics were assigned to each team and spanned 14 weeks. Each reflection contained responses to four compulsory questions:

- *What problem are you trying to solve?*
- *What is your design solution?*
- *What steps did you take?*
- *What would you do differently next week?*

Additionally, students were requested to answer at least one of the following questions:

- *What did you learn about the problem this week?*
- *How did the solution evolve?*
- *What steps did you perform this week?*
- *What went well?*
- *What didn't go well?*
- *What could you have done to improve the outcome?*
- *Were there any limitations you encountered this week?*

The primary purpose of these reflections was to encourage critical thinking and assist students in resolving organizational and technical challenges faced in their team. A secondary purpose was to provide instructors with information to support course delivery and to facilitate targeted interventions when teams encountered difficulties or were progressing slower than others. As the class sizes were relatively large for a design setting, this highlighted the need for a system to improve the efficiency of reviewing engineering students' design reflections, enabling rapid and tailored feedback.

Following the institutional research ethics board approved protocol (#6012355), at the end of each year, students were asked for consent to include their reflections in the study. The next sections describe how the model was trained using the reflection data.

3.2 Data Preprocessing

In all modern AI driven processes involving natural language, preprocessing is required to convert raw text into

a representation suitable for computer interpretation. The first step in preprocessing is to clean the data, which follows the procedure summarized in Appendix A. Thereafter, term frequency-inverse document frequency (TF-IDF) was employed to create numeric feature vectors. TF-IDF reveals how unique or informative words are in the corpus, which provides valuable semantic information [23]. Although more advanced word embedding techniques exist, such as BERT and Word2Vec, they are often more nuanced and complex, whereas TF-IDF provides an interpretable and simple implementation suitable for in-house model development and for educators new to AI [24].

Since the user specifies the number of topics for the LDA model to search for in a corpus, rather than arbitrarily defining the number of topics, *k*-means clustering was performed to guide its selection. The optimal number of topics was determined using the elbow method, with the within-cluster sum of squares (WCSS), computed using cosine similarity, as the validation metric [25]. A lower WCSS indicates greater similarity between topics within a cluster, whereas a higher value suggests greater variance. The elbow method identifies the number of clusters that achieves a balance between well-defined cluster separation and diminishing returns from adding additional clusters [25]. The overall methodology is summarized in Figure 1.

3.3 Model Training

To train the model, the data was partitioned into training and testing sets. The training set corresponded to all the data from project 2 and the test set contained all the reflections from project 1. After training, unifying labels were subjectively assigned to each topic.

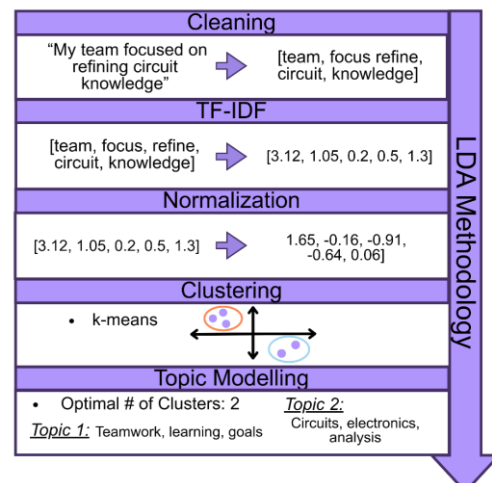


Figure 1. LDA training process.

To quantitatively evaluate the trained model, UMass coherence scores were calculated using the top 30 most representative words from each topic resolved from project

1. The coherence scores provide a statistical measure of how logically and contextually related these words are within each topic. Often, this metric is used to determine how interpretable each topic is to a human. Moreover, values closer to zero often indicate interpretable and semantically similar topics, and larger negative values suggest divergence [26]. Typically, this metric is computed on the training set to reveal if the model was able to group words that are contextually related.

The trained LDA model was also qualitatively evaluated on the testing set through triangulation with the two authors who did not participate in model development. The two authors independently reviewed five sequential reflection sets from the testing set, along with seven randomly selected reflections, totalling 24 reflections. The sequential reflections were randomly selected to ensure representation across gender identities, with three female, three male, and one non-binary student responses. Each sequential set contained between three to five reflections presented in chronological order. Reviewing sequential reflections would help determine whether latent semantic structure exists between weeks that would assist the reviewers in assigning labels. Four of the sampled reflections were chosen from 2021 and three were from 2022. Qualitative analysis is often considered the optimal evaluation method for natural language models [27].

For classification, the authors were provided with the unifying labels and associated topic words extracted from the training data. They were then tasked with assigning the most representative labels to each reflection they were given from the test set. The resulting classifications were triangulated and compared against the model's assignments. The author involved with model development did not evaluate the labels to reduce potential bias. In an ideal scenario, the LDA model's labels would agree with the reviewers' labels, indicating that the themes extracted by the model are representative of the recurrent patterns present within the reflections. In conjunction, this would signify that the model was able to extract meaningful themes while maintaining high-fidelity reference to previous reflection stages, effectively addressing the limitation of many LLM models.

4. FIRST RESEARCH OBJECTIVE: RESULTS

The model exhibited strong quantitative performance but performed poorly under human review. Throughout the process of preparing data, training, and evaluating various model configurations, many of the nuances of AI were elucidated. This led to the conceptualization of the second research objective, shifting the overarching aim from developing a model to review reflections to documenting the potential risks in blindly trusting new tools.

The optimal number of topics identified via clustering was six, as shown in the elbow plot in Figure 2. As an initial

drop in the WCSS was not evident, the elbow was approximated using the distance-to-line method. It was noted that the WCSS decreased monotonically, suggesting that topics became increasingly differentiated as additional clusters were removed, indicating that topics contain discernable thematic information for the model. The most significant changes in WCSS were noted for values less than six, with marginal differences thereafter.

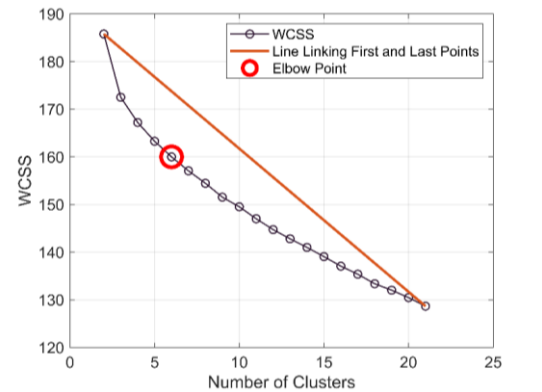


Figure 2. Elbow plot used to determine the optimal number of topics.

The mean coherence score was found to be -1.88, with individual scores for each topic shown in Table 1. Because coherence is not an objective metric, as the range of values and their interpretation depends on the setting, it is common to compare these scores to related work or to other models during development. For instance, Wilson et al., in their study of student perceptions of automated evaluation systems, found an optimal coherence score of approximately -5 and concluded that promising performance could be achieved in extracting student sentiments from topics as they were meaningfully grouped [28]. Our LDA model achieved significantly greater coherence scores. Furthermore, a relatively low degree of variance was observed, with each topic achieving similar scores. Although a more rigorous comparison would be required to determine whether our model performs comparably to that of Wilson et al., this suggests that our model produces meaningful results.

Table 1: UMass coherence score.

Topic	Unifying Label
1	-1.70
2	-1.59
3	-1.85
4	-1.83
5	-2.02
6	-2.25

Table 2 shows the most probable topic words extracted from the model along with the six manually assigned unifying labels/thematics. Interestingly, assigning labels to each topics proved to be a daunting task, as the extracted

words reflected broad terms, many of which carried context dependent semantic meaning such as “think,” “focus,” or “feel.” Additionally, substantial semantic overlap between the words from different topics was observed adding ambiguity to the already subjective process.

Table 2: Reflection themes.

Topic	Unifying Label	Words Extracted from Model from Most to Least Probable
1	In-Class Work and Assignments	Work, time, document, like, think, semester, take, feel, class, due
2	Project Research	Research, group, good, requirement, member, learn, come, lot, find, statement
3	Community Partner Meetings	Team, partner, meeting, community, meet, report, discuss, crate, clinic, visit
4	Project Documentation	Report, final, process, presentation, begin, plan, create, focus, CAD, new
5	Building	Build, material, test, part, prototype, make, building, change, testing, start
6	Problem Empathizing	System, visit, look, site, need, figure, water, able, help, perform

The LDA model was then deployed on the testing data, and topics were assigned to each reflection. The two authors who were manually assigning topics to seven reflections noted a lack of clarity distinguishing the themes, which alerted them to exercise greater caution when interpreting the reflections. Table 3 shows the predicted topic by the LDA model, and the labels assigned by two of the authors. To reduce bias, the third author performed the comparison. Sequential triangulation results are provided in Appendix B.

Table 3: Triangulation between authors and model for seven randomly assigned reflections.

Random Reflection	Predicted Topic	Author #1's Labels	Author #2's Labels
A	1	4	1
B	3	5	2
C	6	4	4
D	1	4	2
E	4	2	2
F	1	1	4
G	4	4	1

* Green cells/bold face indicate agreement between an author and the model's prediction. Orange cells/italics represent agreement between the two authors.

5. DISCUSSION AND IMPLICATIONS

Although the quantitative results were promising, triangulation was not successful between reviewers and the model as shown in Table 3 on the seven randomly sampled reflections. The authors only agreed with each other twice (shown in orange), and there were no observed instances where they both agreed with the model. The ambiguity of the topic words was further exacerbated by low agreement between reviewers and the trained model on both the seven randomly sampled reflections and the sequential sets, with matching labels observed in only seven of the 24 total instances. One reviewer had four concordant labels (16%) and the other had three (12%); however, there were zero-cases where both reviewers simultaneously agreed with the model. In the absence of the model, the reviewers only agreed with each other eight times total on both the random and sequential reflections labels.

This then raised the question: does this indicate the subjectiveness of reviewing reflections, was the review methodology flawed, or could there be a problem with the quantitatively correlated model? The consensus among the three authors is that the answer is nuanced.

5.1 Second Research Objective: Identifying a Framework for AI Learners

The intent of the LDA model was to enable more efficient reflection review, allowing for review on a more regular basis. This would allow educators to study how the content of reflections changed throughout the design process. Though the model performed reasonably well on the quantitative evaluation metrics, the actual themes were not practically meaningful or useful for deployment in the classroom. Statistically, it was significant, but not practical.

It did not appear that the topic words extracted from the LDA model were meaningfully grouped into distinct themes (Table 2). Although the topics may not appear related, individual words may carry latent contextual meaning and are similar only when considering patterns across a much larger subset of reflections. Unlike the human evaluators, the model has access to the full distribution of reflections and can leverage subtle contextual associations and statistical regularities that are not obvious at the individual reflection level. As a result, words may be separable for the model, but their subtleties are nebulous for reviewers. If a human cannot interpret the model's topics, however, the results lose pedagogical significance.

Even though coherence scores are frequently cited as a robust evaluation method to determine topic interpretability by humans, statistical similarity may not always translate to distinct groups in settings such as reflective environments, where complex semantic and affective structures are deeply intertwined [29]. This

highlights how the field of AI is nuanced and context dependent.

One framework, the hierarchy of competence, shown in Figure 3, is a concept that shows how learners gain knowledge in a particular field [30]. This is also called the conscious competence learning model, and can describe how researchers learning to interpret AI models may be misled.

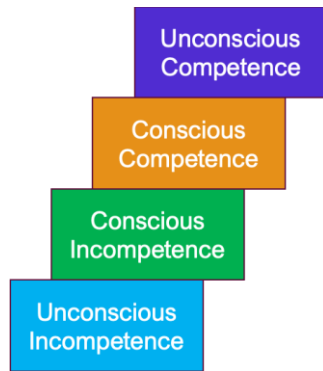


Figure 3. Hierarchy of competence.

In the hierarchy of competence, the learner begins at unconscious incompetence, and they do not know enough about the field or topic to understand their knowledge gaps. We often see this with students who are new to utilizing AI to assist them with assignments for new courses: they cannot evaluate the information that the LLM is providing. As the learner progresses to conscious incompetence, they become more aware of their knowledge gaps. Then the last

two stages switch the order of awareness moving from conscious to unconscious competence. In the third stage, they are still cognitively working through the decisions. The final stage is unconscious competence, where the learner automatically performs the task without conscious effort.

The hierarchy of competence can serve as a methodological framework for learners and engineering educators attempting to use a new instructional tool. As they progress through the hierarchy and become more familiar with the opportunities and cautions with language models, it is important to do this reflectively and thoughtfully.

Connecting the results from Research Objective 1 with the hierarchy of competence [30], because the numerical values demonstrated significance, someone with unconscious incompetence might be tempted to deploy the model for use in the classroom. It successfully output different themes and was quantitatively validated. Without the authors scrutinizing how meaningful the themes are, this could have been unhelpful and even harmful if deployed in a classroom environment.

The humility that comes with being a life-long learner (CEAB graduate attribute 12) should also extend to engineering educators. As we attempt to deploy new tools in the classroom, we must recognize our limitations and realistically assess our abilities. If we are not yet consciously incompetent, that includes getting external support. These observations motivated that adaption of the hierarchy of competence for AI settings, shown in Figure 4.

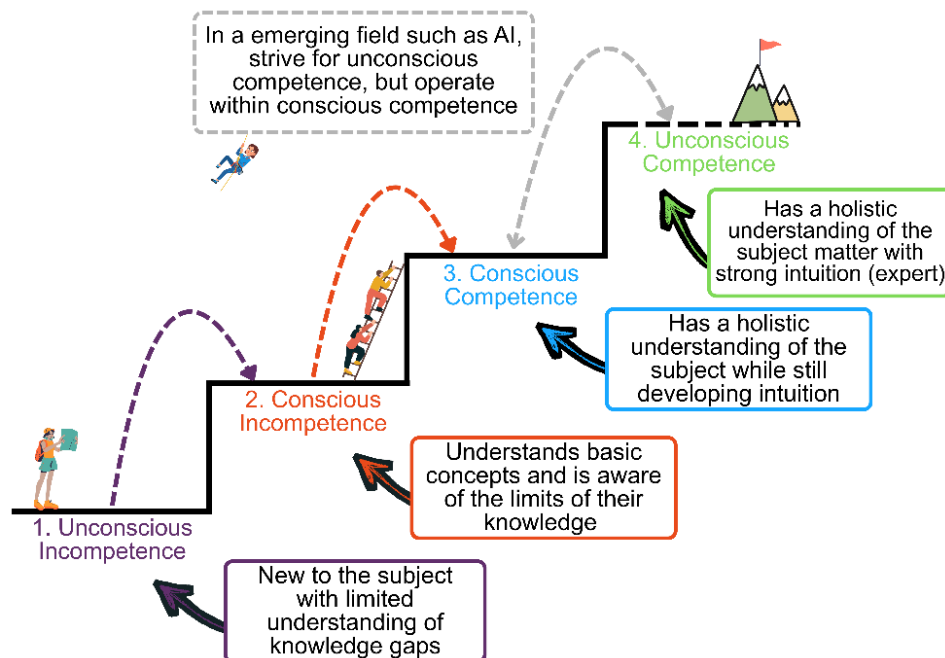


Figure 4. Hierarchy of competence and AI.

Though educators strive for unconscious competence, it is recommended to operate within conscious competence. This is because of the intrinsic nuances of AI coupled with how quickly AI is evolving. As demonstrated with our model, where quantitative results suggested strong performance, even minor contextual variations, sometimes undifferentiable by humans, can result in vastly different outcomes. This is also demonstrated in other AI disciplines, such as vision tasks, where small perturbations to training data can lead to often unexplainable shifts in the output [31].

5.2 Industrial Implications of AI Deployment

As an increasing number of systems rely on AI, proportionally an even greater number of engineers and students are required to learn its fundamentals. In the current dynamic landscape of AI, the authors believe that guidelines governing the use of the professional engineer's seal should be updated to explicitly state that it is the engineers' duty to ensure that AI model performance and reliability are rigorously evaluated before deployment. Numerous studies hypothesize that many AI models published in academia or deployed in industry exhibit overoptimistic results relative to their actual performance [32, 33, 34].

However, beyond that, AI systems are likely to be integrated into systems of systems that are produced by traditional engineering disciplines. Agriculture fields will be watered and weeded by AI powered machines to provide food security. International shipping lanes will be populated by ships where AI is performing most tasks, reducing the risk of loss of life by simply having fewer people on ship. What is the responsibility of engineers in these scenarios? Surely the farmers relying on the AI to improve crop yields for their livelihood and those who work on an AI-powered ship will not be expected to sign an end-user licence agreement!

For software engineers, the world is shifting substantially. Building software systems has long been the hallmark of this discipline, but we are already seeing AI taking on much of the programming task. The role of the software engineer will instead center on communicating with clients, understanding their requirements, deploying AI tools to build and deliver systems. There will likely be a growing emphasis on verification and validation as essential skills, yet these skills are often underrepresented in computer science and software engineering curricula compared to programming and software architecture. The stakes are especially high across safety-critical industries such as aviation, where certification by a credentialed engineer is required to provide rigorous estimation and documentation of risk to human life.

These risk analyses often rely on verification and validation of the software. In specific cases, such as flight control systems, formal verification provides mathematical

proof through a variety of techniques that a system behaves as intended under certain conditions. However, those systems have code that is inspectable, which is not true for AI. AI operates as a black-box model, generating statistical tokens that change with each execution, making formal verification undecidable. This raises difficult questions: How will engineers account for unverifiable AI systems when endorsing a project? How will they account for multiple AIs working in concert? These problems bear little resemblance to traditional software challenges. Instead, the task becomes predicting the behavior of systems, with or without humans, where risk is measured as probabilities.

These, and surely many more, are questions that are essential for engineers to answer as a discipline and a profession for safety, security, sustainability, and many other qualities where AI upends traditional approaches.

5.3 Responsible AI Use in Educational Settings

The experiences in this research with an AI processing very detailed, very nuanced, reflections give some indicators of where and how to use AI in teaching and learning, in particular processing inherently qualitative data. The way the topic words were generated by the machine likely contributed to the limited overlap across the two human researchers and the AI, due to differences in positionality. While on first review, the generated topic words and their groupings looked like they had clear labels, the inter-rater reliability analysis showed that the humans had very different perspectives from the AI model.

For instance, one researcher comes from an engineering background, while another has a human-centred design and computer science background. The two humans have divergent disciplinary histories, lived experiences, and conceptual vocabularies, which shaped how each analyst read the reflections and applied the codes. This is an expected feature of reflexive thematic analysis, where meaning is constructed through the researcher's standpoint [35].

In contrast, the system that generated the topic words, although capable of simulating coding regularities, does not occupy a social context nor engage in ongoing reflexive practice, both of which are integral in qualitative research. As a result, the logic the system used to group words would be expected to diverge from human analyses grounded in lived experience and positional reflexivity. The place this becomes particularly problematic is the unintended consequences of using a system that looked reasonable. For example, if marks were being assigned based on this coding of reflections, it would be radically different from the human instructor. Thus, caution needs to be taken in employing this type of technology. It is possible to be fooled into thinking the machine is doing what we want it to do.

However, what are we to do? Do we return to reading reflections line by line and abandon the use of AI tools? It

seems ill-advised as these tools have the potential to provide insight into student performance in ways that otherwise are tedious, such as getting an aggregate view of the current performance of a class.

We propose that with the advance in power of AI models since this study was conducted, generative models could provide a more robust approach to review reflections due to their ability to encode substantial amounts of natural language as context. Instead of having the machine create codes or themes, have the humans create and document the codebook as in qualitative methods pre-AI. Then, provide enough context about the data being generated, the people who generated it, the social environment in which the data is situated, the positionality of the human researchers who created the codes, and concrete examples of how and where the codes are applied. With this information, there may be sufficient context to automate processing large amounts of qualitative material from students. This could provide better insight across the collection of students with aggregated data and areas of interest where there are deviations. However, it remains unclear what type of information would need to be included for this approach to be successful, if it is possible. Our future studies will investigate this as a possibility.

We leave the debate of whether the AI machine has its own positionality based on its own experiences and training to the reader and other knowledgeable scholars. The authors also encourage others to report and reflect upon unexpected findings and deviations from original objectives, particularly in AI-related work, where quantitatively compelling results may not directly correlate with practical performance; these insights are often equally as valuable as the original research objective, and the recursiveness demonstrates a researcher's ability to be humble and conscious of their ability while seeking to be lifelong learners. For instance, in this work, the limitations encountered in the first objective led to a broader discussion on the consequences of AI misuse in educational settings. Although we should strive to present state-of-the-art results, it is paramount to remain mindful that research is rarely linear, and unexpected outcomes and insights gained are often equally as valuable as the original research objective.

Acknowledgements

We acknowledge that this study was conducted on the traditional lands of the Mi'kmaq peoples and Mohawk nation. We are grateful to them for their stewardship and for the ability to visit and live on their territory.

We identify as one woman and two men. We are of European descent and document this positionality statement to normalize the practice. One of the authors was an undergraduate student when the work was conducted who was interested in how machine learning can be useful

for engineering educators. The other two authors are educators (engineering and computer science), one of whom is open to examining its efficacy but is cautious about its implementation. The other eagerly pursues its boundaries and advances.

All procedures performed in this study involving human participants were in accordance with the ethical standards of the Research Ethics Board per #6012355.

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Disclosure Statement

Artificial Intelligence Tool: Microsoft Copilot (University of Prince Edward Island institutional instance, GPT 5.2) and OpenAI's ChatGPT. Writing and Editing: Some sections had grammar and readability improved after being drafted by the research team. Research Governance: All conceptualization, data collection and data analysis were undertaken without AI assistance, except where noted as part of the methodology.

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Appendix A: Reflection Data Cleaning Process

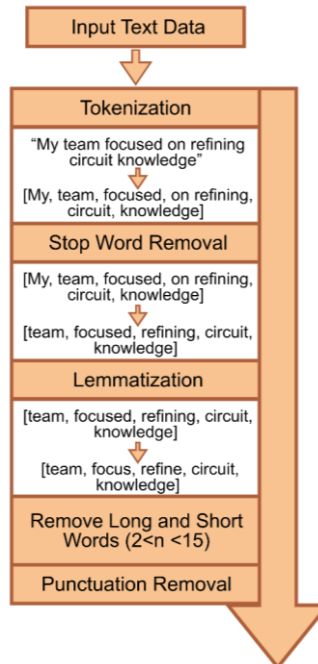


Figure A: Reflection data cleaning process.

Appendix B: Triangulation Results on the Sequential Reflections

To show triangulation alignment, Table B contains the sequential reflections. Green cells/boldface indicate agreement between an author and the model's prediction. Orange cells/italics represent agreement between the two authors. Additional alignment is shown in Table 3.

Table B: Predicted topics versus topics assigned independently by two authors not involved in model development on the randomly sampled sequential reflections.

Sequential Reflection	Sequential Reflection Model Scrutiny														
	Predicted Topics					Author #1's Labels					Author #2's Labels				
	W1	W2	W3	W4	W5	W1	W 2	W3	W 4	W5	W1	W 2	W3	W 4	W5
H	4	4	1	1	6	2	2	2	1	1	4	2	2	2	4
I	4	4	1	1	1	2	2	2	1	1	6	4	3	2	4
J	3	4				2	<i>1</i>				2	<i>1</i>			
K	4	4	4	1	4	3	6	1	4	4	6	2	4	4	1
L	4	4				2	2				2	2			